Lecture 2 - 01/08/2025

Consumption: (Keynes)
↳ $C_t = a + b\, Y_t + e_t$

Exercise (Microdata):
↳ $S_{RATE} = \dfrac{r_{NY} - rent - nd}{r_{NY}}$

→ $S_{RATE} = 1 - \dfrac{C}{Y} = \dfrac{1 - (a + bY)}{Y} = (1-b) - \dfrac{a}{Y}$
↳ same Y → same S_{RATE}

→ black var ⇒ higher savings rate (reality)
↳ change in b or a to explain in Keynes
↳ suboptimal

- Friedman's Permanent Income Theory
  ↳ savings is a way to keep smooth consumption profile in face of fluctuating income
- Modigliani - Bromberg Life-Cycle Theory
  ↳ savings is a way to smooth consumption in face of unsmooth profile over life-cycle

A simple 2-period model:

$\hookrightarrow \max_{c_1, c_2, a_2, \cancel{a_3}} \quad u(c_1) + \beta E u(c_2)$

s.t. $a_{t+1} + c_t = y_t + a_t(1+r)$

assume $a_1, a_3 = 0$

$\rightarrow c_1 = y_1 - a_2$

$\rightarrow c_2 = a_2(1+r) + y_2$

Euler:

$$\underbrace{u'(c_1)}_{\text{MC of save}} = \underbrace{\beta(1+r) E u'(c_2)}_{\text{MU of saving}}$$

Motives for Saving:

$\hookrightarrow$ Consumption - Tilting Motive

$\rightarrow$ non-certainty, $y_1 = y_2$

$\hookrightarrow u'(c_1) = \beta(1+r) u'(c_2)$

$\rightarrow$ case I: $\beta(1+r) > 1 \Rightarrow u'(c_1) > u'(c_2) \Rightarrow c_1 < c_2$

$\hookrightarrow$ MU decreasing

$\hookrightarrow$ save in period I

$\rightarrow$ case 2: $\beta(1+r) < 1 \Rightarrow u'(c_1) < u'(c_2) \Rightarrow c_1 > c_2$

$\hookrightarrow$ borrow in period I

$\rightarrow$ true in multi-period model

$\rightarrow \beta > \frac{1}{1+r}$ worthwile to save

$\rightarrow \beta < \frac{1}{1+r}$ not worth saving

$\rightarrow$ key driver is agent patience

$$\max_{c_1, c_2, a_2, \cancel{a_3}} \; u(c_1) + \beta E u(c_2)$$

$$\text{s.t.} \quad a_{t+1} + c_t = y_t + a_t(1+r)$$

assume $a_1, a_3 = 0$

$\rightarrow c_1 = y_1 - a_2$

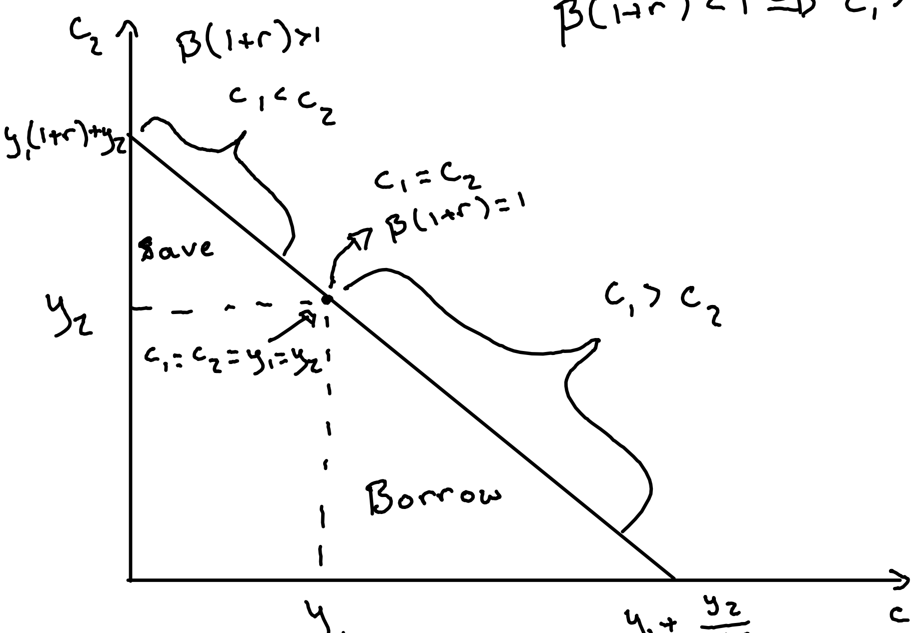
$\rightarrow c_2 = a_2(1+r) + y_2$

$$\Rightarrow \max \; u(c_1) + \beta u(c_2)$$

$$\text{s.t.} \quad c_1 + \frac{c_2}{1+r} = y_1 + \frac{y_2}{1+r}$$

(set $c_i = 0$ to find intercepts)

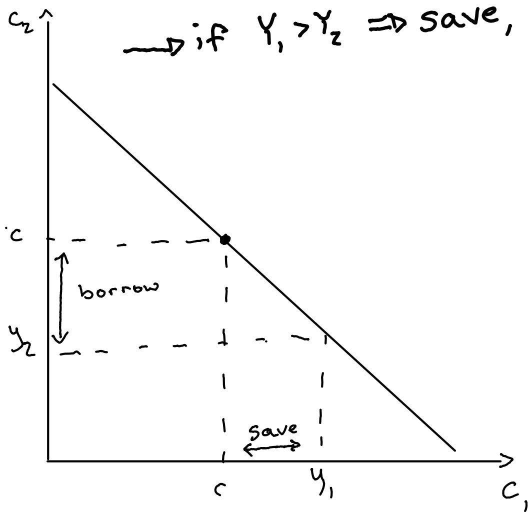
$\beta(1+r) < 1 \Rightarrow c_1 > c_2$



Motives for Saving:

↳ consumption Smoothing Motive:

- → $\beta(1+r)=1$
- → No uncertainty but $Y_1 \neq Y_2$
- → Euler: $U'(c_1) = U'(c_2) \Rightarrow c_1 = c_2$
- → if $Y_1 > Y_2 \Rightarrow$ save, if $Y_2 > Y_1 \Rightarrow$ borrow



Reality check:

↳ CT: implies monotonic consumption profile

↳ CS: implies constant consumption profile (flat-line)

↳ reality: consumption is ∩ shaped

Motives for Saving:

↳ Precautionary Motive:

→ $\beta(1+r)=1$

→ $U'''(\cdot) > 0$ (explain later)

→ $y_1 = E(y_2)$ (expect constant income)

→ $U'(c_1) = EU'(c_2)$ (Euler)

→ consider plan of no saving

↳ $c_1 = y_1$; $c_2 = y_2$

→ $u'(E(y_2)) < EU'(y_2) \Rightarrow$ Jensen Ineq

↳ convex: $E(f(x)) > f(E(x))$

→ concave: opposite

↳ save

→ add uncertainty ⇒ save

Example:

↳ Lottery 1: $c_2 = c_1 + \begin{cases} \frac{c_1}{4} & p = \frac{1}{2} \\ -\frac{c_1}{4} & p = \frac{1}{2} \end{cases}$

↳ lottery 2: $c_2 = c_1 + \begin{cases} \frac{c_1}{2} & p = \frac{1}{2} \\ \frac{c_1}{2} & p = \frac{1}{2} \end{cases}$